MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5693

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, RF sharp cutoff

© Outline --- 8-1 (EIA)

Base --- B8-21, low-loss phenolic

Envelope --- MT8

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	sh	h	g3	g1	k	g2	h	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Pg2	Pp	© Ehk	Ik	TE	© Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	W	W	v	mAdc	°C	ft
Maximum:	6.6	300	-50	125	-100	0.3	2.0	100	10	60	(See
Minimum:	6.0	---	-1 (See note 1)	---	0	---	---	---	---	---	note 2)
TEST CONDITIONS:	6.3	250	-3	100	0	---	---	---	---	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
1331	Variable-frequency vibration	Rp = 10,000 ohms	---	---	Ep	---	40	mVac
	Quality conformance inspection, part 1		©					
1256	Total grid current	Rg1 = 5 Meg (see note 3)	0.65	II	Ic1	0	-0.1	µAdc
1231	Emission current	Eb = Ec1 = Ec2 = Ec3 = 30 Vdc (see note 3)	0.65	II	Is	85	---	mAdc
1306	Transconductance		0.65	II	Sm	1,400	1,900	µmhos
1246	Audio frequency noise	Rk = 3,200 ohms; Rg1 = 0.25 Meg; Rg2 = 1.0 Meg; Rp = 0.25 Meg; Ebb = Ecc2 = 225 Vdc; Esig = 29 mVac	0.65	II	EB	---	17	vu
1256	Electrode current (1) (anode)		---	---	Ib	2.3	3.7	mAdc
1256	Electrode current (2) (anode)	Ec1 = -7.5 Vdc	---	---	Ib	2	80	µAdc
1256	Electrode current (screen)		0.65	II	Ic2	0.6	1.1	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 10,000 ohms	---	---	Ep	---	40	mVac
1256	Electrode current (3) (anode)	Ec3 = -70 Vdc	---	---	Ib	150	750	µAdc
1301	Heater current		---	---	If	290	310	mA
© 1336	Heater-cathode leakage		---	---	Ihk	---	5	µAdc
1211	Insulation of electrodes	See note 4	4.0	S3	R	300	---	Meg
1331	Direct-interelectrode capacitance	Without shield	---	---	Cgp	---	0.005	pF
		Without shield			Cin	4.8	5.8	pF
		Without shield			Cout	5.6	6.8	pF
© 1041	Shock	540 G; Ehk = +100 Vdc (see note 5)	15	S2	---	---	---	---
1031	Vibration fatigue		---	---	---	---	---	---
---	Post-shock and vibration-fatigue test end points	Low-frequency vibration	---	---	Ep	---	100	mVac
		Transconductance	---	---	Sm	1,250	2,050	µmhos
		Total grid current	---	---	Ic1	---	-0.5	µAdc
		Heater-cathode leakage	---	---	Ihk	---	20	µAdc
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; Ehk = 100 Vdc; Eb = 300 Vdc; Ecc2 = 300 Vdc; Rg2 = 90,000 ohms; Rk = 300 ohms; Rg1 = 2 Meg (see note 6)	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours)	Transconductance	---	---	Sm	1,300	---	µmhos
1506	Heater-cycling life	Ef = 7.5 V; Ehk = 100 Vdc; t = 1-1/2 min "on", 1-1/2 min "off" (see note 7)	---	---	---	---	---	---
---	Heater-cycling life test end point	Heater-cathode leakage	---	---	Ihk	---	10	µAdc

NOTES:

1. Minimum negative grid bias = 0.5 Vdc permissible in resistance coupled amplifier service.
2. See "Reduced pressure (altitude) voltage breakdown characteristics in basic document.
3. This test is to be performed at the conclusion of the holding period.
4. Tie shell to ground.
5. Metal inserts shall be used for shock mounting.
6. Intermittent heater voltage; 120 minutes "on", 20 minutes "off". At the conclusion of the life test, the average life of the life-test group of tubes shall not be less than 475 hours.
7. The sample size for heater-cycling life test is determined on the basis of Group A life-test sampling. For lot acceptance no defects are permitted in a sample size of 10 tubes or less. More than one defect on the first sample is cause for rejection of the lot. If one defect occurs in the first sample an additional sample shall be tested. The combined sample size shall be 25 tubes. More than one defect in the combined sample is cause for rejection of the lot. However, if the first sample has one failure, and five preceding heater-cycling life tests showed no defects, the product may be shipped without additional testing.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, MU
Navy -
Air Force - 11, 85
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2401-45)

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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U.S.Army Electronics Command
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-E-1/81C, ELECTRON TUBE, RECEIVING TYPE 5693

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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